Jun 2024 exam: full worked solutions

Practice questions · Day 21 · Fri 13 Nov · pack MX2

Question 1. AC and DC Power Circuit Analysis [20 marks]

1a) Fill-In-The Blank Questions [4 marks]

i) A transformer has an input voltage of 240V and an output voltage of 24V. Based on this it has a turns ratio n of _____ and is a step _____ (choose down or up) transformer. Another transformer has an input voltage of 240V and an output voltage of 480V. Based on this it has a turns ratio n of _____ and is a step _____ (choose down or up) transformer.

ii) An electric motor has an input power of 1000W and a power loss of 300W. This results in an output power of _____W and an efficiency of _____%. Increasing the power loss for the same input power would cause the efficiency to _____ (choose decrease or increase). Increasing the input power for the same power loss would cause the efficiency to _____ (choose decrease or increase).

MARKS: 8 blanks at ½ mark each.

> **COMMON MISTAKE:** writing 1:10 for the first ratio. The course writes primary:secondary, so 240 V to 24 V is 10:1. The other mistake is answering “decrease” for the last blank. Think of it as the loss staying fixed while the input grows.

1b) Electrical and Mechanical Power [3 marks]

Consider a 3ph, 4 pole, 50 Hz induction motor. Under a certain operating condition, the motor speed is 1420 rpm, input voltage is 415 V, input current is 6.8 A, input electrical power is 4 kW and output torque is 23 Nm.

i) Find its input power factor (you can assume this is lagging). [1 mark]

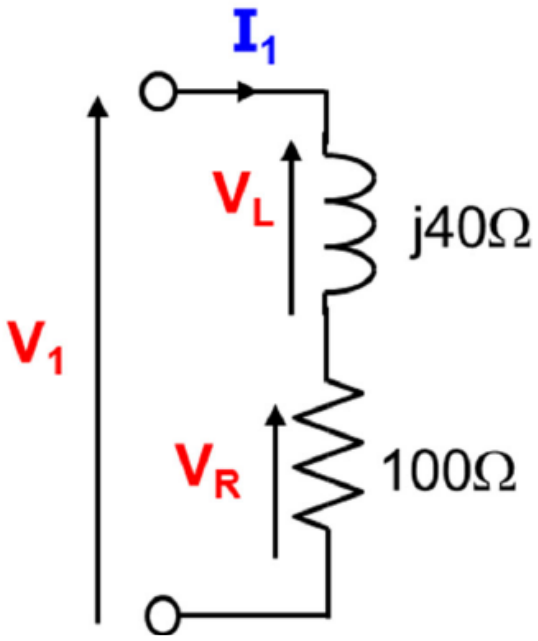
ii) Find its output power and hence efficiency. [2 marks]

MARKS: (i) 1 for $\sqrt{3}V_L I_L$ and pf. (ii) 1 for $P = \tau\omega$ with ω in rad/s, 1 for η.

COMMON MISTAKE: leaving out $\sqrt{3}$, which gives pf = 1.417. A pf above 1 is impossible, so this should trigger a re-check. The second classic error is $P = \tau \times n$ with n in rpm.

1c) AC Circuit Analysis [5 marks]

Consider the series resistor-inductor (RL) circuit shown on the right.

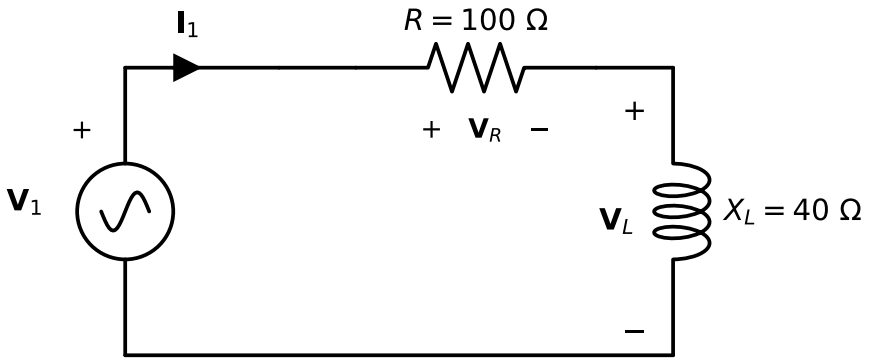


Jun24 Q1c figure: series j40 Ω and 100 Ω carrying I1, with V_L, V_R and V1 marked (from the exam paper)

i) The input current I_1 has a magnitude of 2A. Find the value of the real power and reactive power absorbed by the resistor. Find also the real and reactive power absorbed by the inductor. [3 marks]

ii) For a different value of input current I_1, it is known that the resistor voltage V_R has a magnitude of 50V and the inductor voltage V_L has a magnitude of 20V. Hence find the magnitude of the input voltage V_1. [2 marks]

series RL; values from the paper's figure



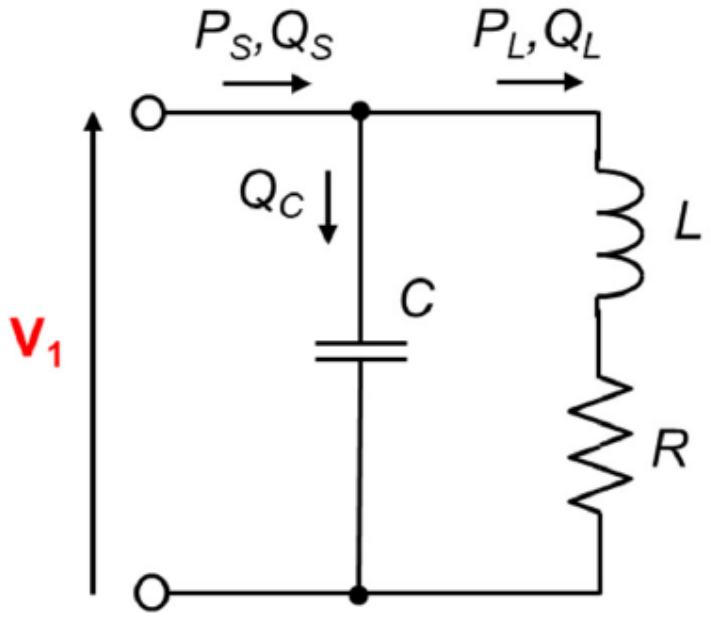
Q1c: series R = 100 ohm and XL = 40 ohm, as in the paper's figure

MARKS: (i) ½ mark for each of the four values, plus 1 for showing I^2R and I^2X (3). (ii) 1 for the perpendicular-phasor reasoning, 1 for 53.85 V.

COMMON MISTAKE: adding magnitudes, $50 + 20 = 70$ V. Also, in (i), using the source voltage instead of the current, or giving the inductor a non-zero P.

1d) Power Factor Correction [8 marks]

Consider the circuit shown on the right which contains a series RL load which is known to absorb a real power of 80W and a reactive power of 60VAr. A power-factor correction capacitor C is connected across the input supply.

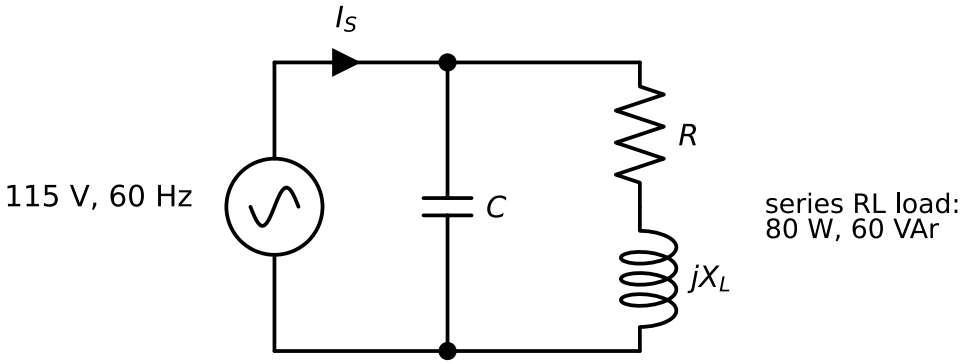


Jun24 Q1d figure: capacitor C across V1 in parallel with the series L and R load; P_S, Q_S at the supply, P_L, Q_L into the load, Q_C into C (from the exam paper)

i) Sketch the power triangle for the RL load and hence calculate its power factor. [2 marks]

ii) By adding the power factor correction capacitor C, it is desired to have a supply (input) power factor of 0.9 lag at the input to the circuit shown. Find the supply real power P_S and the value of supply reactive power Q_S which gives the desired power-factor. Hence find the value of the reactive power absorbed by the capacitor Q_C. [4 marks]

iii) For another desired value of power-factor, the capacitor is found to absorb -30VAr. What is the required value of capacitance (in µF) for the above capacitor if the input power supply is 115V, 60 Hz? [2 marks]



Q1d: capacitor C across the 115 V, 60 Hz supply in parallel with the series RL load (80 W, 60 VAr)

MARKS: (i) 1 for the labelled triangle, 1 for 0.8 lag. (ii) 1 for P_S, 1 for Q_S with a correct $\tan$, 1 for the Q balance, 1 for Q_C with its sign. (iii) 1 for the formula, 1 for the value.

COMMON MISTAKE: $Q_C = +21.25$ VAr (wrong sign). Also using −21.25 VAr in (iii), which gives 4.262 µF. Using 50 Hz gives 7.220 µF. The question says 60 Hz.

Question 2. Magnetic Circuits and Transformers [20 marks]

2a) Electromagnetics [7 marks]

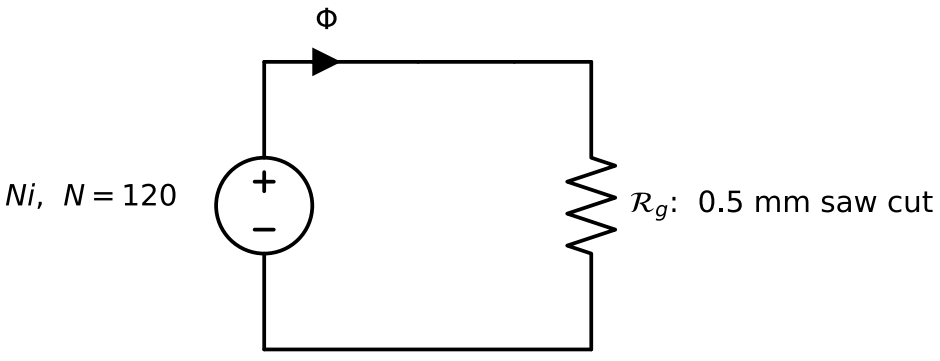
Consider a toroidal magnetic core which has a circular cross-section whose diameter is 2.5cm. The core has a winding of 120 turns and can be assumed to have an infinite permeability. A fine saw is used to cut a thin slot to produce an airgap of 0.5 mm in one side of the core.

i) Calculate the reluctance of the airgap $\mathcal{R}_g$ and hence the inductance of the winding. [3 marks]

ii) If the maximum allowable peak core flux density is 1.5T, calculate the maximum 50Hz voltage which can be applied to the winding. [2 marks]

iii) Using a flux density of 1.5T, calculate the force between the sides of the slot in the core. [2 marks]

toroid, $\mu = \infty$ (core reluctance zero), circular section 2.5 cm diameter



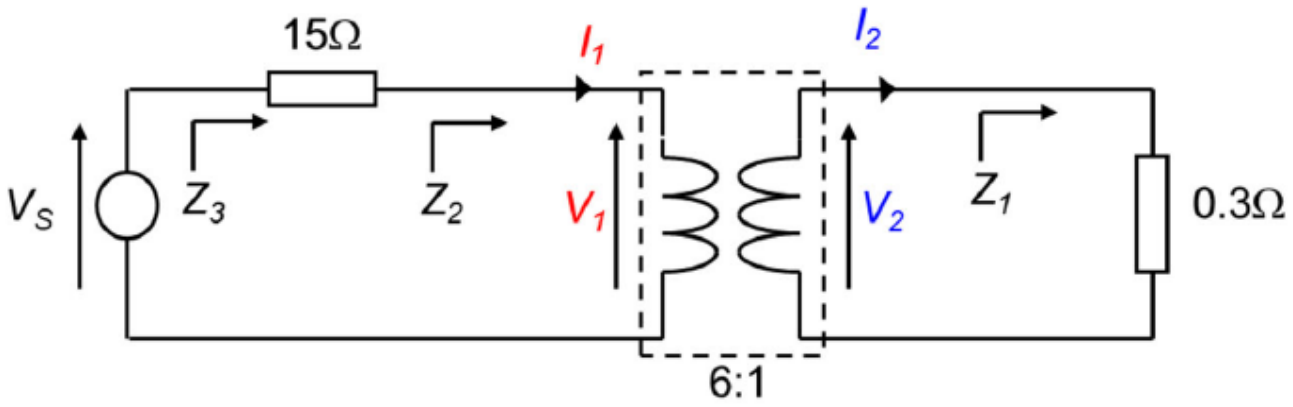
Q2a: magnetic circuit, MMF Ni with N = 120 across the gap reluctance only

MARKS: (i) 1 for the area, 1 for $\mathcal{R}_g$, 1 for L. (ii) 1 for the back-emf equation, 1 for the value. (iii) 1 for the formula, 1 for the value.

COMMON MISTAKE: using the diameter as the radius (A four times too big). Also writing $V = \hat{B}$-limited from L and a current, which needs the reluctance and goes wrong. In (iii), using $B = 1.5/\sqrt{2}$: the force formula uses the stated flux density, 1.5 T.

2b) Ideal Transformers [5 marks]

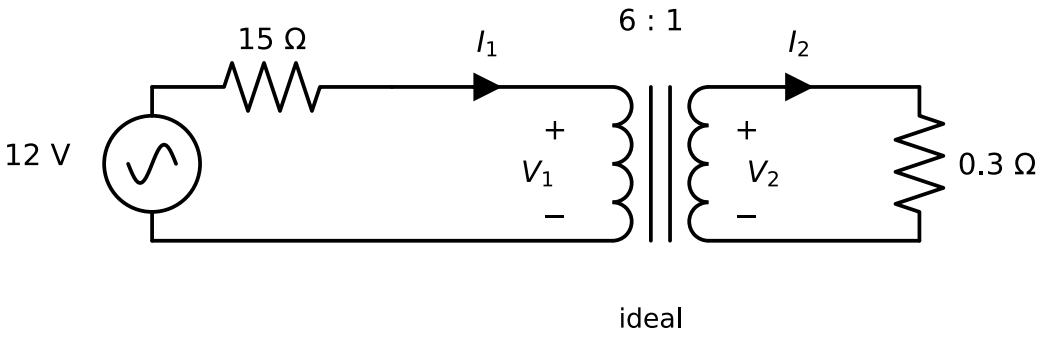
In the diagram on the right the transformer is ideal and has a turns ratio of 6:1.



Jun24 Q2b figure: source V_S, 15 Ω series resistor, 6:1 ideal transformer and 0.3 Ω load, with Z3, Z2, Z1, I1, I2, V1, V2 marked (from the exam paper)

i) Find the values of the impedances Z_1, Z_2 and Z_3 shown in the figure. [2 marks]

ii) Assuming the AC supply voltage V_S is 12V, find the magnitudes of the primary current I_1 and the actual (not referred) secondary current I_2. Find also the magnitudes of the primary voltage V_1 and the actual secondary voltage V_2. [3 marks]



ideal

values from the paper's figure

Q2b: 12 V source, 15 ohm series resistor, 6:1 ideal transformer and 0.3 ohm load

MARKS: (i) 1 for $Z_2 = n^2 Z_1$, 1 for Z_3. (ii) 1 for I_1, 1 for I_2, 1 for both voltages.

> **COMMON MISTAKE:** referring the wrong way $(0.3/36)$, or using $I_2 = I_1/n$. A step-down transformer gives a larger secondary current. Also $V_1 = 12$ V: the 15 Ω resistor drops most of the source voltage.

2c) Three-Phase Transformers [8 marks]

A three-phase transformer is rated at 16 kV/220 kV, 150 MVA. It has the following equivalent circuit parameters (referred to the primary winding where appropriate):

primary resistance R_1 = 0.08 ohms, secondary resistance R_2' = 0.06 ohms
primary leakage reactance X_1 = 0.14 ohms, secondary leakage reactance X_2' = 0.14 ohms
magnetising reactance X_M = 14 ohms, core-loss resistance R_C = 250 ohms.

i) Draw a simplified equivalent circuit for the transformer showing the values for each of the equivalent circuit parameters as impedances. [2 marks]

ii) Consider the transformer in the open-circuit condition with rated input voltage. Find the input current magnitude and (real) power. [4 marks]

iii) Consider the open-circuit condition as discussed above, if the input voltage was halved, by what factor (e.g. x0.25, x0.5, x1, x2 or x4) would you expect the following quantities to change by? [2 marks]

- Input current magnitude: _____

- Input real power: _____
- Input reactive power: _____
- Input power-factor: _____

Q2c: simplified circuit, 250 ohm parallel j14 ohm at the input, then 0.08 + 0.06 ohm and j(0.14 + 0.14) ohm

MARKS: (i) 1 for the shunt branch at the input, 1 for the combined series values. (ii) 1 for $V/\sqrt{3}$, 1 for the parallel Z, 1 for I, 1 for P. (iii) ½ mark for each factor.

> **COMMON MISTAKE:** using 16 kV as the phase voltage, which gives 1145 A and 3.072 MW. Also including the series 0.14 + j0.28 Ω in the OC calculation: on open circuit no current flows in it. And answering pf ×0.5: it does not change.

Question 3. Electric Machines [20 marks]

3a) Torque vs. Speed Characteristics [6 marks]

Consider the torque versus speed curves shown in the figure below.

i) For each curve, write the type of machine in the space below the curve and draw the resulting torque curve if the no-load/synchronous speed was reduced to half of its original value. [3 marks]

ii) Alongside each curve, indicate how you could halve the no-load/synchronous speed while ensuring safe machine operation. [3 marks]

Jun24 Q3a figure: three torque–speed curves, top to bottom a vertical line, a falling straight line and an induction-machine curve (the paper labels the torque axis T) (from the exam paper)

MARKS: (i) 1 per machine for the correct name and halved curve. (ii) 1 per machine. For the AC machines the mark needs both V and f.

COMMON MISTAKE: "halve f" alone for the AC machines. The question says "safe", and the mark is for keeping V/f constant. Another error is drawing the halved DC line through the origin, or with a different slope.

3b) Induction Machine Parameters [4 marks]

The following tests are performed on a 415V, 50Hz three-phase induction motor which is star-connected.

- Stator resistance measurement: 2 ohms per phase
- No-load test: input voltage 415V, input current 3.5A, input power 270W
- Blocked-rotor test: input voltage 80V, input current 5A, input power 390W

Find the following simplified equivalent circuit series parameters: R_1, R_2' and X.

Q3b test circuits: no-load 415 V, 3.5 A, 270 W; blocked rotor 80 V, 5 A, 390 W; R1 = 2 ohm

MARKS: 1 for per-phase values, 1 for R_1, 1 for R_2', 1 for X.

COMMON MISTAKE: not dividing 80 V by $\sqrt{3}$ (this gives $X = 15.13\ \Omega$). Also halving 2 Ω: it is already a per-phase stator value. And using the no-load test for the series parameters.

3c) Induction Machine Calculation [10 marks]

A 415 V, 50 Hz, 960 rpm, three-phase, star-connected induction machine has the following equivalent circuit parameters (referred to the stator winding where appropriate):

stator resistance R_1 = 2.5 ohms, rotor resistance R_2' = 3 ohms
stator leakage reactance X_1 = 6 ohms, rotor leakage reactance X_2' = 6 ohms

The magnetising reactance and core loss resistance can be neglected.

i) The above induction machine is run as a generator at 1025 rpm. Find the synchronous speed, the value of slip s and hence the effective rotor resistance R_2'/s. [3 marks]

ii) Draw a single-phase simplified equivalent circuit for the machine when operating at 1025 rpm, showing the parameter values as impedances. Assuming that the machine is operating at rated voltage, find the magnitude and phase of the input phase current. Hence find the input power. You should assume that the phase angle of the phase voltage is zero. [5 marks]

iii) Using the results from part (ii) find the corresponding airgap power P_{AG} using the fact that in this case $I_2' = I_1$, and hence find the output torque τ. [2 marks]

Q3c: series circuit 2.5 ohm, j6 ohm, j6 ohm and a negative R2′/s, no shunt branch

MARKS: (i) 1 each for n_S, s, R_2'/s. (ii) 1 for the labelled circuit, 1 for $V_1 = 415/\sqrt{3}$, 1 for Z, 1 for I_1 with its angle, 1 for P_{IN} with its sign. (iii) 1 for P_{AG}, 1 for τ (using ω_S).

COMMON MISTAKE: dropping the minus signs ($s = +0.025$ gives a motor at 975 rpm, which is a different machine). Also using $X = 6\ \Omega$ instead of $X_1 + X_2' = 12\ \Omega$, or dividing P_{AG} by the rotor speed instead of ω_S. An angle near ±180° is correct here, not an error.

Question 4. Synchronous Machines [20 marks]

4a) Synchronous Machines [5 marks]

A 14 kV, 400 MVA synchronous generator is tested at a field current of 250 A. The open-circuit voltage was found to be 12 kV and the short-circuit current was 6.5 kA. The stator resistance was measured at 0.2 ohms per phase.

i) Find the value of the synchronous reactance in per-unit. [3 marks]

ii) The synchronous generator is delivering 200MW at a terminal voltage of 3% below rated voltage and a power-factor of 0.8 lagging. Find generator output voltage and current phasors in per-unit. [2 marks]

Q4a test circuits: OC 12 kV line and SC 6.5 kA, both at a field current of 250 A, R = 0.2 ohm

OC and SC characteristics with the Q4a test point at 250 A

MARKS: (i) 1 for $\sqrt{3}$ and $|Z_S|$, 1 for removing R, 1 for Z_B and the pu value. (ii) 1 for $\mathbf{V}$, 1 for $\mathbf{I}$ with the angle.

COMMON MISTAKE: forgetting $\sqrt{3}$ (gives 3.745 pu), or leaving R in (gives 2.175 pu). In (ii), using $|I| = 0.625$ pu, which ignores the 0.97 pu voltage. Answering in ohms also loses the mark: the question asks for pu.

4b) Synchronous Machines [6 marks]

A round-rotor synchronous generator has a synchronous reactance of 1.8 per-unit. Its excitation system is controlled so that its terminal voltage is always 1 per-unit. Resistance and saturation may be neglected.

i) What excitation (in per unit) is required to produce the generator's rated volt-amp output at 0.9 leading power-factor? [3 marks]

ii) What is the rotor angle at this loading? [1 mark]

iii) If both the terminal voltage and the excitation are fixed at the above levels, what is the theoretical maximum power, in per-unit, that can be delivered without losing synchronism? What is the rotor angle corresponding to this condition? [2 marks]

Q4b: generator E0 behind j1.8 pu, V = 1 pu, current out at rated VA and 0.9 pf leading

MARKS: (i) 1 for $\mathbf{I}$ with the $+$ angle, 1 for $\mathbf{E}_0 = \mathbf{V} + jX_S\mathbf{I}$, 1 for 1.634 pu. (ii) 1 for δ. (iii) 1 for P_{max}, 1 for 90°.

COMMON MISTAKE: using -25.84° (lagging). That gives $E_0 = 2.410\angle 42.23^\circ$ pu and loses the whole part. Also recomputing E_0 in (iii): "excitation fixed" means keep 1.634 pu and set $\delta = 90^\circ$.

4c) Short-Answer Questions [9 marks]

Answer the following questions using words only (that is, without using any sketches, drawings or equations).

These model answers are built from the official answer sheet. The sheet says you do not need every point for full marks. Aim for one solid point per mark.

4c(i) [3 marks]. We have discussed three types of electric power: real, reactive and apparent. Explain briefly what each one is and their practical significance.

MARKS: 1 per power type, for its definition plus its practical significance.

> **COMMON MISTAKE:** writing $S = P + jQ$ or drawing a triangle. The question forbids equations and sketches. Another is saying reactive power "is wasted". It is not consumed. It oscillates, and its cost is the extra current.

4c(ii) [3 marks]. Consider a transformer operating from an AC supply. Carefully explain what are the main types of losses in the transformer, how are they modelled in the simplified equivalent circuit, and how do these vary with output power.

MARKS: 1 for the copper loss mechanism and model, 1 for the iron loss mechanism and model, 1 for how each varies with output.

> **COMMON MISTAKE:** putting the core loss in series, or saying iron loss rises with load. Also forgetting that the question asks "how do these vary": give both trends explicitly.

4c(iii) [3 marks]. Carefully explain why the input current of an induction machine under no-load conditions is not zero. What are the physical effects which cause this?

MARKS: 1 for magnetising current and X_M with the physical reason, 1 for core loss and R_C, 1 for mechanical losses (friction and windage) and the small slip.

> **COMMON MISTAKE:** saying "the current is zero because there is no load torque". Also listing only the equivalent-circuit elements without the physical cause: the answer sheet asks for both.